

Project Name: Capstone D-03

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BILL OF MATERIAL

Controller

	Reference TM221ME16R/G Description TM221ME16R (screw), TM221ME16RG (spring) 8 digital inputs, 8 relay outputs (2 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 24 Vdc power supply controller with removable terminal blocks. Power supplied to the IO bus 5V: 520 mA / 24V: 460 mA
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Modules

	Reference TM3DQ8R/G Description TM3DQ8R (screw), TM3DQ8RG (spring) 8-channel, 2A relay outputs expansion module with 2 common lines and removable terminal block. Consumption on the IO bus 5V: 25 mA / 24V: 40 mA
	Reference TM3AI8/G Description TM3AI8 (screw), TM3AI8G (spring) Expansion module with 8 analog inputs (+- 10 V, 0-10V, 0-20 mA, 4-20mA), 12 bits, removable terminal blocks. Consumption on the IO bus 5V: 35 mA / 24V: 0 mA
	Reference TM3AI8/G Description TM3AI8 (screw), TM3AI8G (spring) Expansion module with 8 analog inputs (+- 10 V, 0-10V, 0-20 mA, 4-20mA), 12 bits, removable terminal blocks. Consumption on the IO bus 5V: 35 mA / 24V: 0 mA

HARDWARE CONFIGURATION

MyController - TM221ME16R/G

Digital Inputs

Used	Address	Filtering	Latch	Run/Stop	Events	Priority	Subroutine
	%I0.0	3 ms			Not Used		
	%I0.1	3 ms			Not Used		
	%I0.2	3 ms			Not Used		
	%I0.3	3 ms			Not Used		
	%I0.4	3 ms			Not Used		
	%I0.5	3 ms			Not Used		
	%I0.6	3 ms			Not Used		
	%I0.7	3 ms			Not Used		

Digital Outputs

Used	Address	Status Alarm	Fallback value	Used by
X	%Q0.0		0	User logic
X	%Q0.1		0	User logic
X	%Q0.2		0	User logic
X	%Q0.3		0	User logic
X	%Q0.4		0	User logic
X	%Q0.5		0	User logic
X	%Q0.6		0	User logic
X	%Q0.7		0	User logic

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
	%IW0.0	0 - 10 V	Normal	0-1000	0	
	%IW0.1	0 - 10 V	Normal	0-1000	0	

Fast Counters

Used	Address	Input	Configured	Preset	Double Word
	%FC0	%I0.2	NotUsed	0	
	%FC1	%I0.3	NotUsed	0	
	%FC2	%I0.4	NotUsed	0	
	%FC3	%I0.5	NotUsed	0	

High Speed Counters

Used	Address	Type
	%HSC0	Not Configured
	%HSC1	Not Configured
	%HSC2	Not Configured
	%HSC3	Not Configured

ETH1

Device name:	M221
IP Mode:	Fixed
IP address:	192.168.100.195
Subnet mask:	255.255.255.0
Gateway address:	192.168.100.1
Transfer Rate:	Auto
Security Parameters:	Programming protocol enabled
	Auto discovery protocol enabled
	Modbus server enabled
	EtherNet/IP protocol enabled

Modbus TCP

Unit ID:	247
Output registers (%IWM):	10
Input registers (%QWM):	10

SL1 (Serial line)

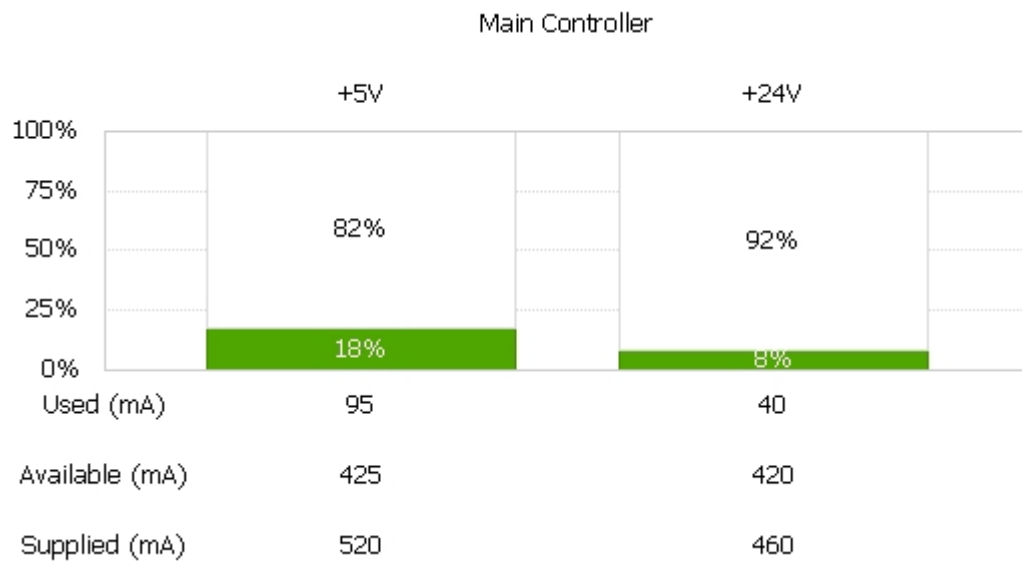
Physical Settings

Device:	TMH2GDB
Baud rate:	19200
Parity:	Even
Data bits:	8
Stop bits:	1
Physical medium:	RS-485
Polarization:	No

Protocol Settings

Protocol:	TMH2GDB
Response timeout (× 100 ms):	10
Time between frames (ms):	10
Transmission mode:	RTU
Addressing:	Slave
Address:	1

IO Bus



TM3DQ8R/G

Digital Outputs

Used	Address	Fallback value
X	%Q1.0	0
X	%Q1.1	0
X	%Q1.2	0
X	%Q1.3	0
X	%Q1.4	0
X	%Q1.5	0
X	%Q1.6	0
X	%Q1.7	0

TM3AI8/G

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
	%IW2.0	Not used	Not used	0-0	0	1 ms/Channel
	%IW2.1	Not used	Not used	0-0	0	1 ms/Channel
	%IW2.2	Not used	Not used	0-0	0	1 ms/Channel
	%IW2.3	Not used	Not used	0-0	0	1 ms/Channel
	%IW2.4	Not used	Not used	0-0	0	1 ms/Channel
	%IW2.5	Not used	Not used	0-0	0	1 ms/Channel
	%IW2.6	Not used	Not used	0-0	0	1 ms/Channel
	%IW2.7	Not used	Not used	0-0	0	1 ms/Channel

TM3AI8/G

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
	%IW3.0	Not used	Not used	0-0	0	1 ms/Channel
	%IW3.1	Not used	Not used	0-0	0	1 ms/Channel
	%IW3.2	Not used	Not used	0-0	0	1 ms/Channel
	%IW3.3	Not used	Not used	0-0	0	1 ms/Channel
	%IW3.4	Not used	Not used	0-0	0	1 ms/Channel
	%IW3.5	Not used	Not used	0-0	0	1 ms/Channel
	%IW3.6	Not used	Not used	0-0	0	1 ms/Channel
	%IW3.7	Not used	Not used	0-0	0	1 ms/Channel

SOFTWARE CONFIGURATION

Constant Words

KW

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

KD

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

KF

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

Network Objects

Input Assembly (Ethernet/Ip)

Used	Address	Symbol	Fallback value	Comment
------	---------	--------	----------------	---------

Output Assembly (Ethernet/Ip)

Used	Address	Symbol	Comment
------	---------	--------	---------

Input Registers (Modbus Tcp)

Used	Address	Symbol	Fallback value	Comment
------	---------	--------	----------------	---------

Output Registers (Modbus Tcp)

Used	Address	Symbol	Comment
------	---------	--------	---------

Digital inputs (IOScanner)

Used	Address	Channel	Symbol	Comment
------	---------	---------	--------	---------

Digital outputs (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment
------	---------	---------	----------------	--------	---------

Input registers (IOScanner)

Used	Address	Channel	Symbol	Comment
------	---------	---------	--------	---------

Output registers (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment
------	---------	---------	----------------	--------	---------

Software Objects

Timers

Allocation: Automatic
Allocated: 0

Counters

Allocation: Automatic
Allocated: 0

LIFO/FIFO Registers

Allocation: Automatic
Allocated: 0

Drums

Allocation: Automatic
Allocated: 0

Shift Bit Registers

Allocation: Automatic
Allocated: 0

Step Counters

Allocation: Automatic
Allocated: 0

Schedule Blocks

Allocation: Automatic
Allocated: 0

RTC

PID

Used	PID	Symbol	Type	Comment
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Grafcet Steps

Allocation: Automatic
Allocated: 0

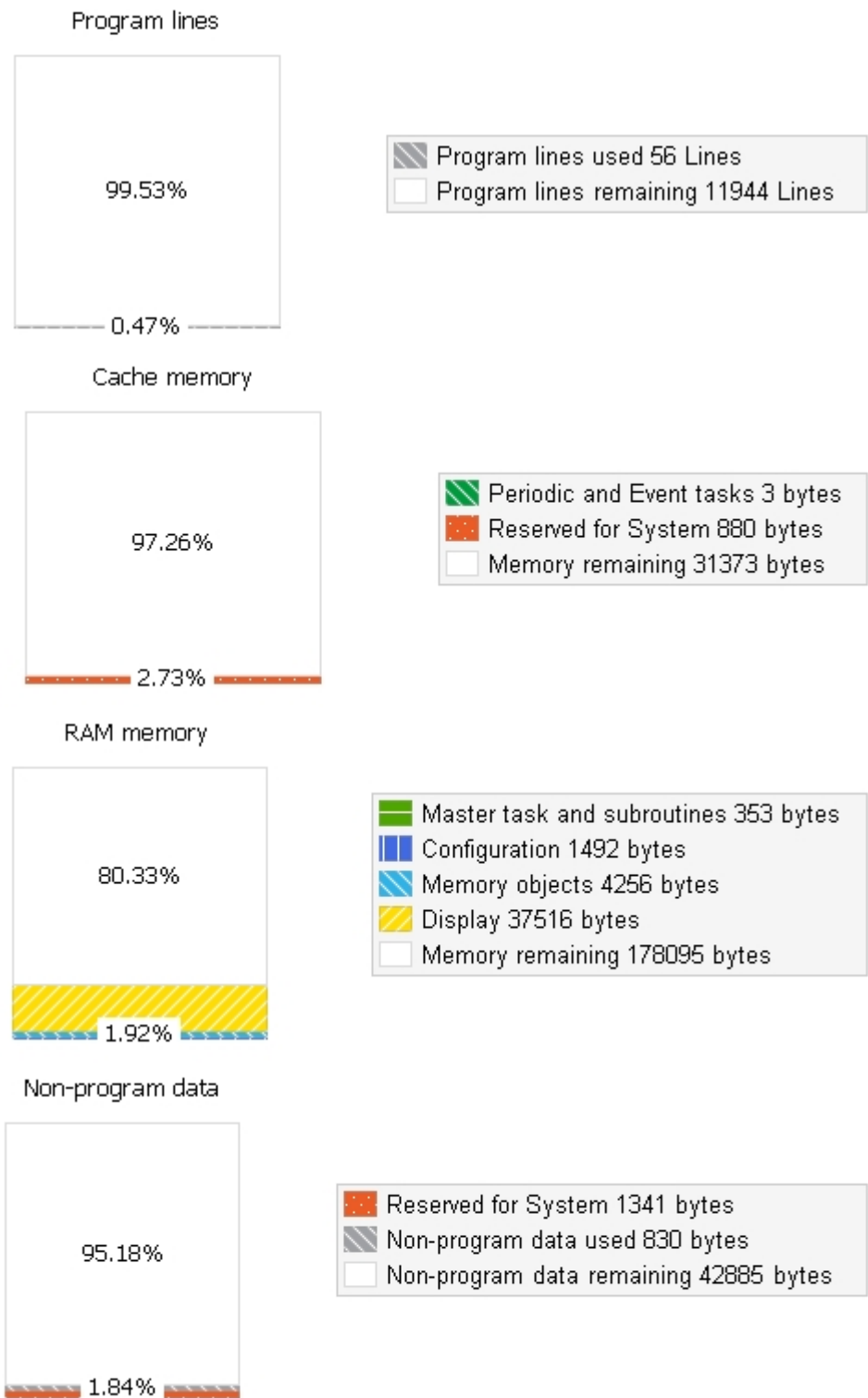
PROGRAM

Behavior

Functional level:	Level 11.0
Starting mode:	Start In Previous State
Watchdog:	250 ms
Fallback behavior:	Fallback value
String end character:	CR (Carriage Return)

Memory consumption

Last compilation: 25/05/2026 21:00:49



Application Architecture

Master Task

Scan mode: Normal
POU list: 1 - New POU

Periodic Task

Period: 255 ms

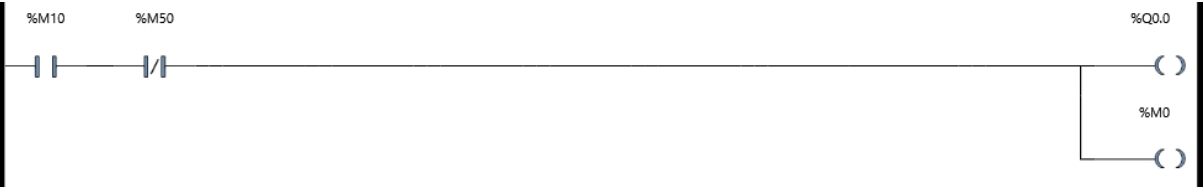
POU

Master Task

1 - New POU

Master Task

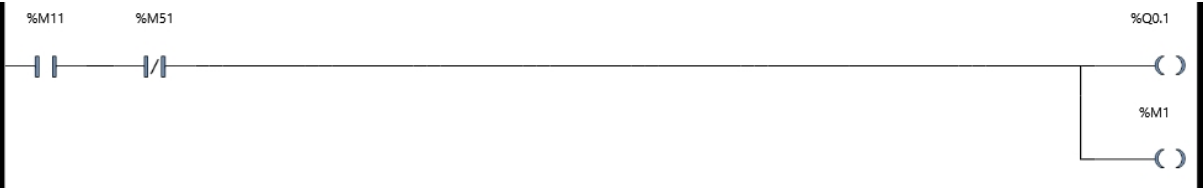
Rung0 - Gen 1A: 75 MW



Variables used:

%M0	READ_GEN1
%M10	STATUS_GEN_1A_75MW
%M50	CONTROL_GEN1
%Q0.0	COIL_GEN1

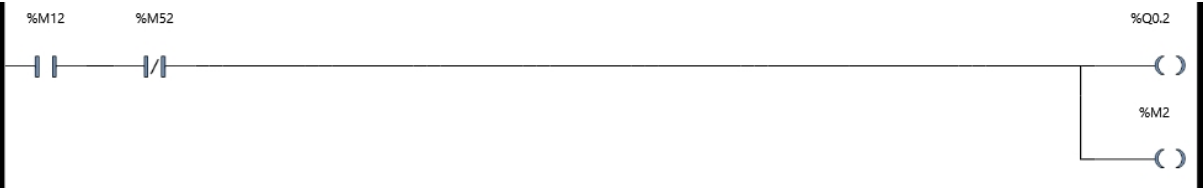
Rung1 - Gen 1B: 25 MW



Variables used:

%M1	READ_GEN2
%M11	STATUS_GEN_1B_25_MW
%M51	CONTROL_GEN2
%Q0.1	COIL_GEN2

Rung2 - Gen 2A: 75 MW



Variables used:

%M2	READ_GEN3
%M12	STATUS_GEN_1B_75MW
%M52	CONTROL_GEN3
%Q0.2	COIL_GEN3

Rung3 - Gen 2B: 25 MW

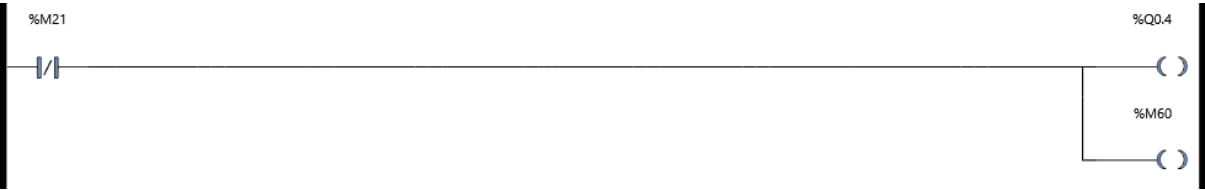


Variables used:

%M3	READ_GEN4
%M13	STATUS_GEN_2B_25_MW
%M53	CONTROL_GEN4
%Q0.3	COIL_GEN4

Rung4 - L101

Comment: Prioritas 2

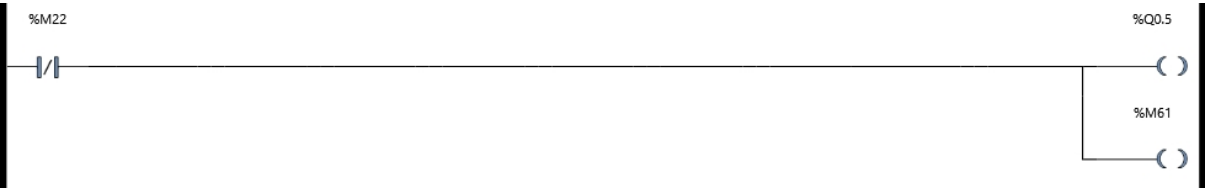


Variables used:

%M21	L101
%M60	READ_LOAD1
%Q0.4	COIL_LOAD1

Rung5 - L201

Comment: Prioritas 2

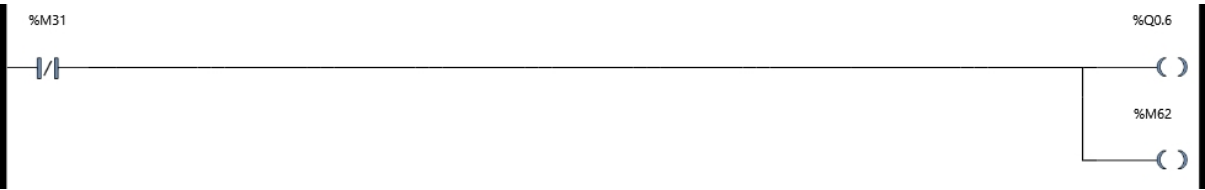


Variables used:

%M22	L201
%M61	READ_LOAD2
%Q0.5	COIL_LOAD2

Rung6 - L301

Comment: Prioritas 3

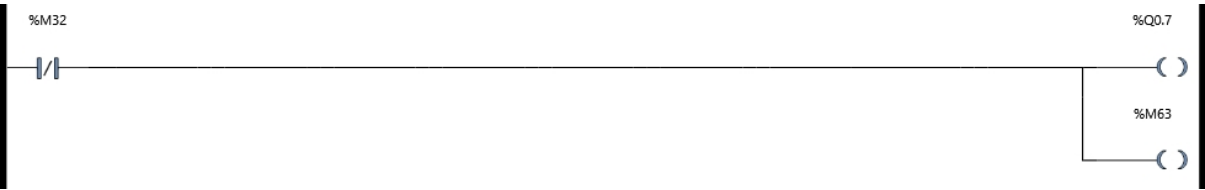


Variables used:

%M31 L301
%M62 READ_LOAD3
%Q0.6 COIL_LOAD3

Rung7 - L302

Comment: Prioritas 3

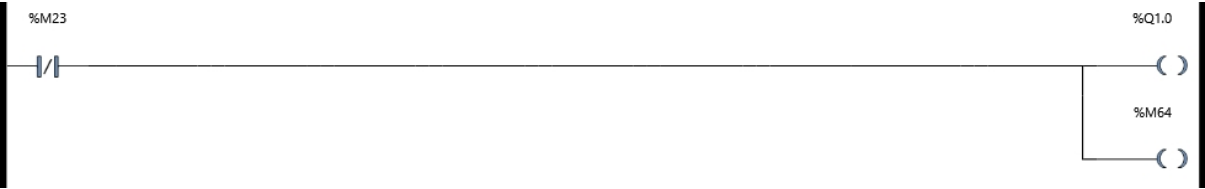


Variables used:

%M32 L302
%M63 READ_LOAD4
%Q0.7 COIL_LOAD4

Rung8 - L303

Comment: Prioritas 2

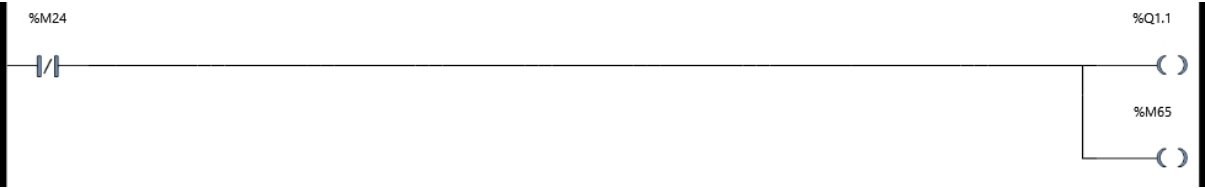


Variables used:

%M23 L303
%M64 READ_LOAD5
%Q1.0 COIL_LOAD5

Rung9 - L304

Comment: Prioritas 2

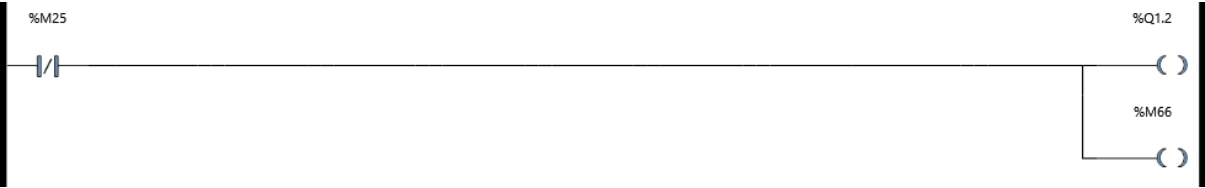


Variables used:

%M24	L304
%M65	READ_LOAD6
%Q1.1	COIL_LOAD6

Rung10 - L305

Comment: Prioritas 2

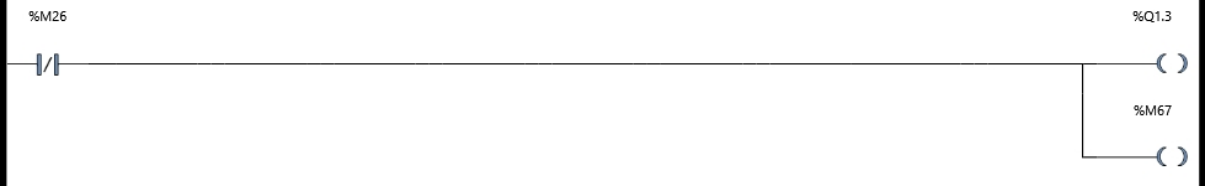


Variables used:

%M25	L305
%M66	READ_LOAD7
%Q1.2	COIL_LOAD7

Rung11 - L401

Comment: Prioritas 2

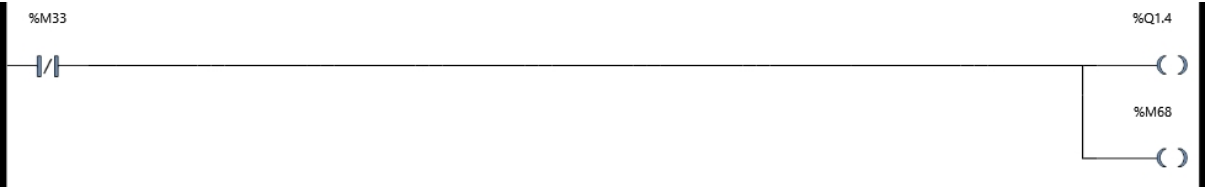


Variables used:

%M26	L401
%M67	READ_LOAD8
%Q1.3	COIL_LOAD8

Rung12 - L402

Comment: Prioritas 3

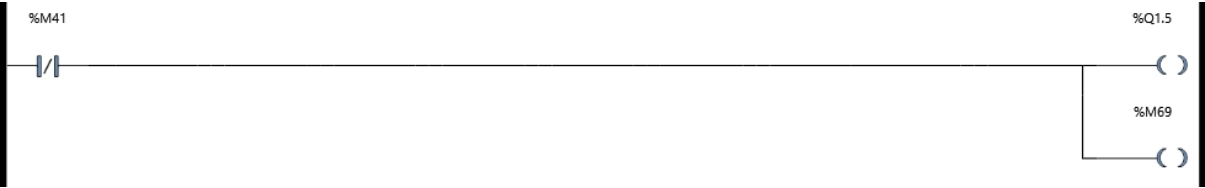


Variables used:

%M33	L402
%M68	READ_LOAD9
%Q1.4	COIL_LOAD9

Rung13 - L403

Comment: Prioritas 4

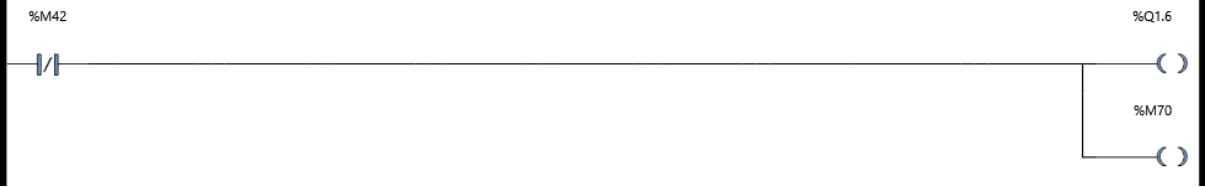


Variables used:

%M41	L403
%M69	READ_LOAD10
%Q1.5	COIL_LOAD10

Rung14 - L404

Comment: Prioritas 4

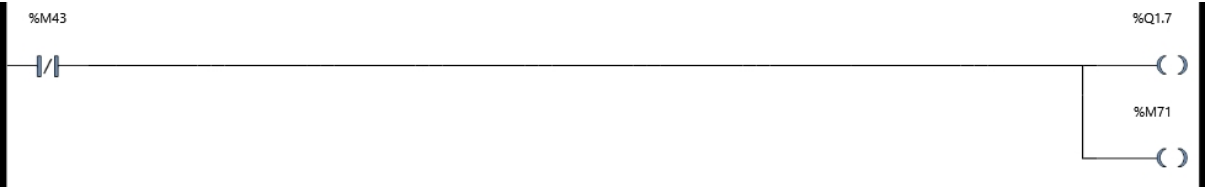


Variables used:

%M42	L404
%M70	READ_LOAD11
%Q1.6	COIL_LOAD11

Rung15 - L405

Comment: Prioritas 4



Variables used:

%M43	L405
%M71	READ_LOAD12
%Q1.7	COIL_LOAD12

DISPLAY

General Properties

Date format: dd/mm/yyyy

Time format: 24 hh/mm/ss

Password protect Setup: Yes

Password: 6301

Operator Interface home page: Setup menu (112)

F1 Key:

F2 Key:

Pages

SYMBOLS

Used	Address	Symbol	Comment
X	%M0	READ_GEN1	
X	%M1	READ_GEN2	
X	%M2	READ_GEN3	
X	%M3	READ_GEN4	
X	%M10	STATUS_GEN_1A_75MW	
X	%M11	STATUS_GEN_1B_25_MW	
X	%M12	STATUS_GEN_1B_75MW	
X	%M13	STATUS_GEN_2B_25_MW	
X	%M21	L101	
X	%M22	L201	
X	%M23	L303	
X	%M24	L304	
X	%M25	L305	
X	%M26	L401	
X	%M31	L301	
X	%M32	L302	
X	%M33	L402	
X	%M41	L403	
X	%M42	L404	
X	%M43	L405	
X	%M50	CONTROL_GEN1	
X	%M51	CONTROL_GEN2	
X	%M52	CONTROL_GEN3	
X	%M53	CONTROL_GEN4	
X	%M60	READ_LOAD1	
X	%M61	READ_LOAD2	
X	%M62	READ_LOAD3	
X	%M63	READ_LOAD4	
X	%M64	READ_LOAD5	
X	%M65	READ_LOAD6	
X	%M66	READ_LOAD7	

Used	Address	Symbol	Comment
X	%M67	READ_LOAD8	
X	%M68	READ_LOAD9	
X	%M69	READ_LOAD10	
X	%M70	READ_LOAD11	
X	%M71	READ_LOAD12	
X	%Q0.0	COIL_GEN1	
X	%Q0.1	COIL_GEN2	
X	%Q0.2	COIL_GEN3	
X	%Q0.3	COIL_GEN4	
X	%Q0.4	COIL_LOAD1	
X	%Q0.5	COIL_LOAD2	
X	%Q0.6	COIL_LOAD3	
X	%Q0.7	COIL_LOAD4	
X	%Q1.0	COIL_LOAD5	
X	%Q1.1	COIL_LOAD6	
X	%Q1.2	COIL_LOAD7	
X	%Q1.3	COIL_LOAD8	
X	%Q1.4	COIL_LOAD9	
X	%Q1.5	COIL_LOAD10	
X	%Q1.6	COIL_LOAD11	
X	%Q1.7	COIL_LOAD12	

CROSS-REFERENCE TABLE

Address	Object	Rung	Code
%M0.....	1 - New POU	Rung0 - Gen 1A: 75 MW	--()--
%M1.....	1 - New POU	Rung1 - Gen 1B: 25 MW	--()--
%M2.....	1 - New POU	Rung2 - Gen 2A: 75 MW	--()--
%M3.....	1 - New POU	Rung3 - Gen 2B: 25 MW	--()--
%M10.....	1 - New POU	Rung0 - Gen 1A: 75 MW	-- --
%M11.....	1 - New POU	Rung1 - Gen 1B: 25 MW	-- --
%M12.....	1 - New POU	Rung2 - Gen 2A: 75 MW	-- --
%M13.....	1 - New POU	Rung3 - Gen 2B: 25 MW	-- --
%M21.....	1 - New POU	Rung4 - L101	-- / --
%M22.....	1 - New POU	Rung5 - L201	-- / --
%M23.....	1 - New POU	Rung8 - L303	-- / --
%M24.....	1 - New POU	Rung9 - L304	-- / --
%M25.....	1 - New POU	Rung10 - L305	-- / --
%M26.....	1 - New POU	Rung11 - L401	-- / --
%M31.....	1 - New POU	Rung6 - L301	-- / --
%M32.....	1 - New POU	Rung7 - L302	-- / --
%M33.....	1 - New POU	Rung12 - L402	-- / --
%M41.....	1 - New POU	Rung13 - L403	-- / --
%M42.....	1 - New POU	Rung14 - L404	-- / --
%M43.....	1 - New POU	Rung15 - L405	-- / --
%M50.....	1 - New POU	Rung0 - Gen 1A: 75 MW	-- / --
%M51.....	1 - New POU	Rung1 - Gen 1B: 25 MW	-- / --
%M52.....	1 - New POU	Rung2 - Gen 2A: 75 MW	-- / --
%M53.....	1 - New POU	Rung3 - Gen 2B: 25 MW	-- / --
%M60.....	1 - New POU	Rung4 - L101	--()--
%M61.....	1 - New POU	Rung5 - L201	--()--
%M62.....	1 - New POU	Rung6 - L301	--()--
%M63.....	1 - New POU	Rung7 - L302	--()--
%M64.....	1 - New POU	Rung8 - L303	--()--
%M65.....	1 - New POU	Rung9 - L304	--()--
%M66.....	1 - New POU	Rung10 - L305	--()--

Address	Object	Rung	Code
%M67.....	1 - New POU	Rung11 - L401	-- () --
%M68.....	1 - New POU	Rung12 - L402	-- () --
%M69.....	1 - New POU	Rung13 - L403	-- () --
%M70.....	1 - New POU	Rung14 - L404	-- () --
%M71.....	1 - New POU	Rung15 - L405	-- () --
%Q0.0....	1 - New POU	Rung0 - Gen 1A: 75 MW	-- () --
%Q0.1....	1 - New POU	Rung1 - Gen 1B: 25 MW	-- () --
%Q0.2....	1 - New POU	Rung2 - Gen 2A: 75 MW	-- () --
%Q0.3....	1 - New POU	Rung3 - Gen 2B: 25 MW	-- () --
%Q0.4....	1 - New POU	Rung4 - L101	-- () --
%Q0.5....	1 - New POU	Rung5 - L201	-- () --
%Q0.6....	1 - New POU	Rung6 - L301	-- () --
%Q0.7....	1 - New POU	Rung7 - L302	-- () --
%Q1.0....	1 - New POU	Rung8 - L303	-- () --
%Q1.1....	1 - New POU	Rung9 - L304	-- () --
%Q1.2....	1 - New POU	Rung10 - L305	-- () --
%Q1.3....	1 - New POU	Rung11 - L401	-- () --
%Q1.4....	1 - New POU	Rung12 - L402	-- () --
%Q1.5....	1 - New POU	Rung13 - L403	-- () --
%Q1.6....	1 - New POU	Rung14 - L404	-- () --
%Q1.7....	1 - New POU	Rung15 - L405	-- () --